

Embedding Earth

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A Critical Atlas of Planetary Computation

experiments in mapping
writing
coding

What does it mean
to predict
embed
simulate
know
query
the Earth?

Remote sensing foundation models are trained on satellite, aerial, radar, thermal, and climate data, to learn — or predict — spatial-temporal representations of Earth's surface.

Earth embeddings construct latent space: a raster image painted by the brushstrokes of a billion vectors.

Massive amounts of Earth observation data are compressed into a unified digital representation, or embedding.

Applications include clustering, crop type mapping, classification, and object detection via similarity search.

How can humans make sense of this not-for-human world?

I conceive of embedding space as
a world not for humans
without clouds
where no one lives

My research exists in the irony between calling for a human understanding of Earth embeddings and acknowledging that attempting to see like machines is a losing game.

Universalizing the planet in a topological space enables us to quantify similarity — or embedding distance — of any time and place on Earth.

everything is relative
locality is universal
distance is semantic

sensing is deterministic
places are queried like the web

I am not so interested in what geospatial embedding data can tell us about the Earth as what this methodology tells us about ourselves.

"The information refers to, but does not simply represent, the space it maps: it exceeds, transforms, and reorganizes that space into another space. Not a representation of a space, but a space itself."

— Laura Kurgan, Close Up At A Distance

Globally sampled 64d AlphaEarth Embeddings

"Our brains are dulled by the incurable mania that consists in reducing the unknown to what is known, to what can be filed."
— André Breton, Manifesto of Surrealism

"The accumulation, storage, and management of information in today's big data repositories raises important questions that have long been at the heart of cultural theories of the archive."
— Uncertain Archives: Approaching the Unknowns, Errors, and Vulnerabilities of Big Data

"When embeddings are used to present and explain research, they take the epistemic role of a map."
— de Bodt et al, Low-Dimensional Embeddings of High-Dimensional Data

"We might be lured into a funhouse of ever-receding and anachronistically enfolding simulacra, where 'real nature' disappears entirely."
— Astrida Neimanis, Nature Represents Itself

Globally sampled 64d AlphaEarth embeddings reduced to 2d with t-SNE

"The globe is on our computers. No one lives there."
— Gayatri Chakravorty Spivak, Imperatives to Re-Imagine the Planet

"Abstraction today is no longer that of the map, the double, the mirror or the concept. Simulation is no longer that of a territory, a referential being or a substance. It is the generation by models of a real without origin or reality: a hyperreal."
— Jean Baudrillard, Simulacra and Simulation

"Do not mistake the availability of data as permission to remain at a distance."
— Yanni Loukissas, All Data Are Local

Globally sampled 64d AlphaEarth embeddings reduced to 2d with UMAP

0.098
-0.207
0.089
-0.154
0.005
-0.200
0.059
-0.025
-0.252
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-0.027
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-0.002
-0.125
-0.207
-0.125
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0.119
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-0.186
0.059
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-0.038
0.142
-0.025
-0.009
-0.020
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